

Feiyang Huang

PhD Candidate, Computational Biology & Medicine

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Education

PhD	Weill Cornell Graduate School of Medical Sciences , Tri-Institutional PhD Program in Computational Biology & Medicine • Advisor: Wesley Tansey, Memorial Sloan Kettering Cancer Center	New York, NY Aug 2023 – present
BS	Johns Hopkins University , Computer Science and Biomedical Engineering • GPA 3.96 • Activities: President, Singapore Students' Association (2021–2023); Theta Tau Professional Engineering Fraternity (2020–2022)	Baltimore, MD Sept 2020 – May 2023

Experience

Memorial Sloan Kettering Cancer Center, Tansey Lab , PhD Researcher Bayesian and generative models for cancer biology and therapeutic design. • Contributed to PPC , a pan-cancer atlas of ex vivo drug screens for functional precision oncology. • Contributed to a population-scale study of clinical and genetic determinants of systemic-therapy toxicity.	New York, NY June 2024 – present
Memorial Sloan Kettering Cancer Center, Tansey Lab , PhD Rotation Bayesian active learning to nominate combination therapies from ex vivo drug screens; groundwork for the PPC atlas.	New York, NY Mar 2024 – May 2024
Memorial Sloan Kettering Cancer Center, Chodera Lab , PhD Rotation Graph neural networks for small-molecule property prediction. • Integrated the ViSNet equivariant GNN into mtenn (model conversion, configuration, tests) and set up CI against stable and nightly PyTorch Geometric.	New York, NY Jan 2024 – Mar 2024
Memorial Sloan Kettering Cancer Center, CMO Innovation Lab , PhD Rotation Computational pipeline for bisulfite- and enzyme-based DNA methylation sequencing of tumor samples, with Brian Loomis.	New York, NY Nov 2023 – Jan 2024
Sloan Kettering Institute, Morris Lab , PhD Rotation BERT model for representation learning from tumor mutation profiles.	New York, NY Aug 2023 – Nov 2023
Johns Hopkins University, Fan Lab , Undergraduate Researcher • Built and benchmarked a reference-free LDA pipeline to deconvolve spatial transcriptomics data (STdeconvolve, <i>Nature Communications</i> 2022). • Developed a torque-based method and simulation framework to quantify subcellular mRNA polarization.	Baltimore, MD Jan 2021 – May 2023
Hologic , Software Engineering Intern • Built an internal work-history search tool and automated release-note generation (3 h to 20 min).	May 2022 – Aug 2022
Johns Hopkins University, Biomedical Design , Team Lead, iMEDS • Led a team of 7 building an algorithm to continuously monitor sedation in pediatric ICU patients; interviewed 20+ clinicians at Johns Hopkins, Boston Children's and Stanford.	Baltimore, MD Aug 2022 – May 2023
Johns Hopkins University, Institute for NanoBioTechnology , Research Intern • Optimized a point-of-care PCR assay (15% lower detection limit) and its embedded image-analysis and motor-control code.	Baltimore, MD May 2021 – Aug 2021
DioTeX Diagnostics , Co-Founder • Led assay development for a lateral-flow test for internal hemorrhage; raised over \$60,000 in non-dilutive funding.	Baltimore, MD Nov 2020 – Aug 2023

National University of Singapore, Yu Lab , Research Intern Trained a CNN for endoscopic image classification (91% accuracy); co-author in <i>Gastrointestinal Endoscopy</i>.	Singapore Nov 2019 – Dec 2020
Singapore Armed Forces , 3rd Sergeant Led a detachment of 3 soldiers and co-supervised daily training for 15.	Singapore Jan 2018 – Oct 2019

Publications

A Pan-Cancer Ex Vivo Drug Screen Atlas for Functional Precision Oncology Karl Pichotta, Jessica B White, Jeffrey F Quinn, Anneliese Markus, Christopher Tosh, Antoine De Mathelin, Erin Coyne, Feiyang Huang , Wesley Tansey pmc.ncbi.nlm.nih.gov/articles/PMC12934811 (bioRxiv (preprint))	Feb 2026
Population-Scale Precision Safety in Oncology Reveals Clinical and Genetic Determinants of Systemic Therapy Toxicity Ziad Bakouny, X Alex Guo, Feiyang Huang , Saksham Mohan, Rohan Walser, et al., Wesley Tansey, Jian Carrot-Zhang, Ed Reznik www.medrxiv.org/content/10.64898/2026.09.16.26363259 (medRxiv (preprint))	Sept 2026
Reference-free cell type deconvolution of multi-cellular pixel-resolution spatially resolved transcriptomics data Brendan F Miller, Feiyang Huang , Lyla Atta, Arpan Sahoo, Jean Fan 10.1038/s41467-022-30033-z (Nature Communications 13, 2339)	Apr 2022
Artificial intelligence-enhanced white-light colonoscopy with attention guidance predicts colorectal cancer invasion depth Xiaobei Luo, Jiahao Wang, Zelong Han, Yang Yu, Zhenyu Chen, Feiyang Huang , et al., Hanry Yu 10.1016/j.gie.2021.03.936 (Gastrointestinal Endoscopy 94(3), 627–638)	Sept 2021
Roles and regulation of long noncoding RNAs in hepatocellular carcinoma Lee Jin Lim, Samuel YS Wong, Feiyang Huang , et al., Caroline G Lee 10.1158/0008-5472.CAN-19-0255 (Cancer Research 79(20), 5131–5139)	Oct 2019

Conference Abstracts

- Bakouny Z, Guo XA, Walser R, **Huang F**, et al. Pan-cancer discovery of clinical and genomic determinants of immune-related adverse events. *Journal of Clinical Oncology* 44(16_suppl):11132 (ASCO 2026).
- Wu S, Pejic B, Roy S, **Huang F**, et al. Development of a computational model of SBS scores in critically ill children. *Pediatric Critical Care Medicine* 25(11S):e87 (2024).

Patents

- Parent C, Simon E, **Huang F**, Ruci A, Zhang E. Methods and devices for quantitatively estimating thrombomodulin. US Patent App. 19/240,058 (2025).
- Eng R, Parent C, Hayat A, Ruci A, Krishnan A, **Huang F**, Simon E, Zhang E. Methods and devices for quantitatively estimating syndecan-1. US Patent App. 17/715,294 (2022).

Presentations

- Military Health System Research Symposium, Kissimmee, FL (2022). Towards point-of-care internal hemorrhage detection through a syndecan-1 lateral flow assay. Oral presentation.
- Johns Hopkins BME Design Day, Baltimore, MD (2022). DioTeX: Hemorrhage Diagnostics. Poster.

Honors and Awards

NIH DEBUT Challenge, Healthcare Technologies for Low-Resource Settings Prize With the DioTeX team; \$15,000.	2023
VentureWell e-Fest Prize With the DioTeX team; \$10,000.	2023

Teaching

- Course Assistant, EN.601.230 Mathematical Foundations for Computer Science, Johns Hopkins University (Fall 2022).

Additional Information

Languages: English, Mandarin